Lab Experiment-7

AIM:-

write a program implement flow contr at data link layer using SLIDING WIN PROTOCOL. simulate the flow of frames f one node to another.

Create a sender program with followin features:-

1. I/P window size from the user
2. I/P a text message from the user
3. consider 1 character per frame.
4. create a frame with following fiel
5. send the frames.
6. wait for the acknowledgement from the receiver.
7. Reader a file called Receiver Buffer.
8. check Ack field for the acknowledgement number.
9. If the Ack number is an expected then send of frames.

Create a receiver file with following features:

```
#include <stdio.h>
#include <string.h>
#define MAX 100
 struct frame {
        int frameNo;
```

```
    char data;
 };
int main(){
     struct frame frames [MAX];
     char message [MAX],
     int totalframes , windowsize,
     int base=0,
     int expectedframe=1;
      Int lostframe;
      int loss simulated=0;
      int i, end;
      int read Retransmission.
     Printf("enter themessage: ");
     fgets(message, sizeof(message), stdin);
    message[strcspn(message,"\n")]='\0',
    total frames = strlen(message);
     if(total frames = =0){
        Printf("Message cannot be empty.\n");
       return 1; }
      Printf("enter window size: ");
    If (scanf("%d",&windowsize>MAX)){
    Printf("Invalid window size.\n");
    return 1; }
  if (window size >total frames){
  window size = total frames; }
   printf("enter frame number tobase
              (0 for none): ");
 If (scanf("%d",& lostframe)!=1|| lostframe<0
                                 || lost frame>total
                                    frames){
```

```
Printf ("invalid frame nomber .\n");
return 1;}
for(i=0; i< totalframes; i++){
frames[i].frameNo=i+1;
frames[i].data=message[i];}
Printf("\n---RECEIVER PROCESS FRAMES...
for(i=base; i<end; i++){
  int frame no == lost frame && !loss
                              simulated){
Printf("Receiver<-frame%d, Data:%c\n"
                   frameNo, frames[i].data);
if (frameNo == lostframe && !lossSimulated){
Printf("Receiver discards act-of-order
        frame %d \n", framer)
 need Retransmission=1;}
Printf ("\n---RECEIVER PROCESS RETRANSMITTED
                        FRAMES\n");
for (i=expectedframe-1; i<end; i++){
 ~~int~~ framer = frame[i].frame;
Printf ("Receiver <-frame%d, Data:%c\n"
                 frameNo, frames[i], data);
if (frameNo == expectedframe){
   Printf("Receiver->ACK %d\n", frameNo);
  expectedframe++; }
 else{
   Printf("Receiver->NACK %d\n", expected
 frame);}}}
```

```
if (base < dataframes) {
    printf("\nwindow slides forward to frame %
            d.\n",
      frames[base].frameno); }
    }
printf("\n- - - - - - - - - - - -\n");
printf("All frames transmitted and acknowledged
                                   \n");
printf("original message: %s\n", message);
printf("= = = = = = = = = = = = = =\n");
return 0;
}
```

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Result::

Thus the study of implement flow control of data link layer using SLIDING window PROTOCOL simulate the flow of frames from one mode to another is done and verified.